

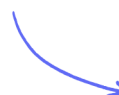
Structured Questions

Alkanes

Alkanes - Introduction / Alkanes as Fuels / Combustion of Alkanes / Alternative Fuels / Free Radicals & Fission / The Free Radical Substitution Mechanism

Medium (1 question)	/15
Hard (1 question)	/5
Total Marks	/20

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Medium Questions

1 (a) Heptane, C_7H_{16} , is an alkane found in crude oil.

Heptane can undergo incomplete combustion.

i) Give a reason why incomplete combustion sometimes occurs.

(1)

ii) Write the equation for the incomplete combustion of heptane, forming carbon monoxide and water as the **only** products. State symbols are not required.

(1)

(2 marks)

(b) Heptane is reformed into branched-chain and cyclic hydrocarbons that are used in petrol.

i) Draw the **skeletal** formulae of a branched-chain alkane and a cycloalkane, each containing **seven** carbon atoms.

(2)

Branched-chain alkane
Cycloalkane

ii) Write the equation for the reforming of heptane into a cycloalkane.

Use molecular formulae.

(1)

iii) Give a reason for adding cycloalkanes to petrol.

(1)

(4 marks)

(c) Heptane, C_7H_{16} , reacts with chlorine in the presence of ultraviolet radiation.

i) State the type and mechanism of this reaction.

(2)

ii) Give the mechanism for the reaction to produce $C_7H_{15}Cl$, $C_{14}H_{30}$ and HCl as the **only** products.

Include the name of each of the steps in your mechanism.
Curly half-arrows are **not** required.

(7)

(9 marks)

Hard Questions

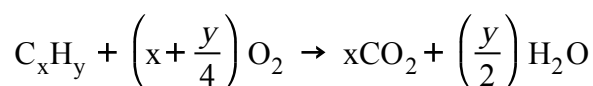
- 1 (a) Using excess oxygen, 25 cm³ of a gaseous hydrocarbon C_xH_y was burned completely.

After cooling to room temperature the total gas volume was measured and found to be 75 cm³ less than the total gas volume before the mixture was ignited.

When the product gases were shaken with potassium hydroxide solution, the total gas volume decreased by a further 100 cm³.

All gas volumes were measured at room temperature and pressure.

A general equation for the combustion of a hydrocarbon is



Determine the molecular formula of C_xH_y. You **must** show your working.

(3 marks)

- (b) When C_xH_y was added to a little bromine water and the mixture shaken, the bromine water remained yellow.

Suggest **two** possible structures for C_xH_y.

(2 marks)